RETAILSHERE (PTY) LTD

Data Platform Programme

Source System Investigation

Step 5 Deliverable — TBC

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Total Systems	**10 source systems investigated**
Status	Draft — Pending CDO Sign-off

1. Purpose

This document presents the findings of Step 5 - Source System Investigation. The purpose of Step 5 is to investigate every source system we discovered in the BDR-001 document and through the stakeholder engagements. Document all the details we will need about each source system, including system overview, data domains served, extraction method to be used, key tables, data quality issues, volume and velocity, and integration complexity.

The findings of this document enable Step 6 ADR resolution, Step 7 dimensional model design, and Step 8 source to target mapping.

2. Scope

This document defines the 10 source systems, including SAP ECC 6.0, Oracle MICROS POS, Shopify Plus, Salesforce CRM, Legacy SQL Server Data Warehouse, Excel Workbooks, Salesforce Marketing Cloud, Meta Business Manager, Google Analytics 4, and Supplier Contract PDFs.

What is explicitly out of scope - dimensional models, source to target mapping, and ADR resolutions are addressed in Steps 6, 7, and 8, respectively.

3. Investigation Framework

Each source system is documented using a seven-field format. This will ensure all information needed for Steps 6, 7, and 8 is captured for every source system, eliminating gaps and rework.

Field	Description
System Overview	What is it? What Version? Who owns it? Who uses it?
Data Domains Served	Which of our 6 domains does this system feed?
Extraction Method	How do we get data out? API? Files Export? Direct DB? Scheduled? Real-time?
Key Tables	What are the most important tables or objects we need?
Data Quality Issues Identified	What problems did we find or anticipate from the interviews?
Volume and Velocity	How much data? How often? Daily Records? Monthly?
Integration Complexity	How hard is this to connect to? Low / Medium / High And why?

4. SAP ECC 6.0

System Overview	SAP ECC 6.0. Core ERP system Business Owner: Johan van der Merwe Technical Owner: TBC — IT/SAP Basis team Serves Procurement, Inventory, and Finance domains. Single point of failure – three domains depend on it simultaneously.
Data Domains Served	Procurement, Inventory Management, and Finance.
Extraction Method	Scheduled nightly export via SFTP in CSV format. Azure Data Factory picks up files from SFTP location and loads into Bronze layer. No direct database connection to production SAP. Extraction window: 11 pm after business closes.
Key Tables	**PROCUREMENT:** EKKO – Purchase Order Header EKPO – Purchase Order Line Items EKBE – Purchase Order History (GRNs)

	RBKP – Invoice Header RSEG - Invoice Line Items **INVENTORY:** MARA – Material Master MARC – Plant level stock MARD – Storage location stock MSEG – Material document segments (Stock movements) **FINANCE:** BKPF – Accounting Document Header BSEG – Accounting Document Items LFA1 – Vendor Master (Suppliers)
Data Quality Issues Identified	**Issue 1 – GRN Timestamp Unreliable** Goods batch-captured at end of shift. SAP records 5pm for goods that arrived at 9am. On-time delivery calculation impossible without reliable timestamps. Bronze layer solution: four timestamp columns per GRN record **Issue 2 – Backdated Postings** SAP allows transactions with past dates. Platform misses postings made after nightly run. Solution: Change Data Capture + alert within 1 hour of detection. **Issue 3 – Manual PO Confirmation** Supplier confirms via email. Buyer manually updates SAP. If update is missed – system shows PO as unconfirmed inaccurately. Solution: Process improvement required. Platform cannot fix manual data entry.
Volume and Velocity	TBC – to be confirmed with SAP basis team during technical investigation. Estimated contribution: 40-50% of total platform daily volume (~300MB overall estimate). Extraction frequency: Nightly batch.
Integration Complexity	MEDIUM Reason 1: Extraction method known – nightly SFTP CSV export. Reason 2: No direct DB connection – no SAP performance impact. Reason 3: Single point of failure – three domain pipelines fail simultaneously if SAP is down. Requires robust retry logic and alerting.

5. Oracle MICROS POS

System Overview	Point of Sale system used across all 87 RetailSphere stores. Records every in-store transaction. Business owner: Rethabile Sithole. Technical owner: TBC – IT/Operations team. Not confirmed. Rethabile Sithole uses the data but does not own the system. Serves Sales and Inventory Management. Risk: 87 stores – 87 different database schemas. Each store might have slightly different table structures, column names, and data types depending on installation date and version. Schema normalisation required before any transformation.
Data Domains Served	Sales and Inventory Management.
Extraction Method	The recommended approach is CDC (Change Data Capture), where the data platform will pull the changed records after the store closes. **TBC items before confirming:** 1. DBMS behind each MICROS store - Oracle, MySQL, or SQL Server? Determines which CDC tool to use. 2. Network connectivity per store - VPN or WAN available to all 87 stores?

	3. Firewall rules - can the platform connect directly to the store DB? 4. IT team confirmation of system ownership and access. **Alternative if CDC is not feasible:** Nightly file export from each store to a central SFTP location. Store generates CSV at close. Platform picks up and processes. Lower complexity. Less real-time.
Key Tables	Transaction Header Transaction Line Items Product Master Employee Store Payment Returns Void
Data Quality Issues Identified	**Issue 1 - Returns and Voids** Customer returns are captured as negative sales in POS, but damaged returns going to the DC are captured separately or not at all. **Issue 2 - No cross-channel customer view** 66% of in-store transactions are completely anonymous. Identity resolution required — ADR-002. **Issue 3 - Slightly different database schema** Each store may have slightly different table structures, column names, and data types depending on installation date and version. Schema normalisation required before any transformation
Volume and Velocity	**Peak estimate:** ~150 transactions per hour per store ~1,500 per store per day ~130,500 across 87 stores per day **Conservative estimate:** ~100 transactions per hour per store ~1,000 per store per day ~87,000 across 87 stores per day Design for peak — 130,500 per day. Size infrastructure for peak load not average load. **Data volume estimate:** Each POS transaction record ~1-2KB 130,500 × 2KB = ~261MB per day from POS alone. **Extraction frequency:** Nightly batch after store close. Or CDC — near real-time if confirmed feasible in the Extraction Method field.
Integration Complexity	HIGH Reason 1: 87 stores — 87 slightly different database schemas. Schema normalisation required before Bronze landing. Reason 2: Connectivity unconfirmed – IT team must confirm firewall rules, network connectivity, and DBMS for all 87 stores. Reason 3: Extraction Method TBC – CDC or file export depends on IT confirmation. Two very different pipeline designs required.

6. Shopify Plus

System Overview	E-commerce platform for RetailSphere online store. Customers purchase products online for home delivery. Business owner: Rethabile Sithole. Technical owner: TBC – IT team confirmation required. Rethabile Sithole uses the sales data but does not own the platform. Serves only the Sales domain. No unified customer identifier – Shopify customer ≠ loyalty member. ADR-002 dependency. No master product identifier – Shopify SKU ≠ SAP stock code. ADR-001 dependency.
Data Domains Served	Sales.
Extraction Method	Method: Shopify REST API, Azure Data Factory has built-in Shopify connector. Schedule: Nightly batch. After business closes. Aligned with SAP and POS extraction window.
Key Tables	Key endpoints to extract: /orders — sales transactions (order date, customer id, product id) /Customers — customer records (customer id, email address, first name, last name, contact number, shipping address) /Products — product catalogue (product id) /Inventory — stock levels (product id, product name, product category)
Data Quality Issues Identified	**Issue 1 — No Master Product Identifier** Shopify SKU ≠ SAP stock code. Same product has different identifiers across systems. ADR-001 must be resolved before product dimension can be built. **Issue 2 — No Unified Customer Identifier** Shopify customer ID ≠ Salesforce loyalty member ID. 66% of loyalty members cannot be linked to their Shopify history. ADR-002 must be resolved. **Issue 3 — Online Returns Captured in POS** Customer returns an online purchase in-store. Return captured in POS not in Shopify. Shopify shows the original sale. POS shows the return. Revenue and stock figures inconsistent across systems.
Volume and Velocity	It would be 5-15% of total retail revenue. So, if in-store is ~87 000-130 000/day, online might be 5-15% of that: ~4 350 to ~19 575 orders per day. **A reasonable mid estimate:** ~5 000 to ~10 000 online orders per day. **Record Size:** Shopify record size is larger than POS - more fields (shipping address, line items, customer details, payment method). Estimate ~3-5KB per order. **Volume Calculations:** 10 000 orders x 4KB = ~40MB per day.
Integration Complexity	LOW-MEDIUM Reason 1: Extraction method known — REST API via ADF built-in Shopify connector. Reduces uncertainty. Reason 2: No direct database connection —API only. No production system performance impact.

	Reason 3: Identity resolution required — Shopify SKU must match SAP stock code (ADR-001). Shopify customer must match loyalty member (ADR-002). Validation rules add complexity in Silver layer.

7. Salesforce CRM

System Overview	It is a Customer Relationship Management system. Has 2.1 million loyalty member records for RetailSphere. Business owner: Priya Naido, Head of CRM and Loyalty. Technical owner: TBC - IT team. It serves the Customer Loyalty and Sales domains. No unified customer ID links loyalty members to POS transactions or Shopify orders. 2.1-million-member records without cross-system resolution. POPIA compliance is required for identity matching. ADR-002 Unified Customer Identity depends on this system as the master customer record. Every other system must link to it as mentioned.
Data Domains Served	Sales and Customer Loyalty.
Extraction Method	Method: Azure Data Factory, Salesforce connector v2 using Copy Activity API: Salesforce Bulk API 2.0 - Designed for large volume extractions — ideal for 2.1-million-member records. Query: SOQL (Salesforce Object Query Language), similar to SQL but specific to Salesforce objects. Schedule: Nightly batch extraction. Aligned with other source system extraction window.
Key Tables and Entities	Contact — loyalty member records (customer ID, email address, address, contact number, first name, and last name) Account — member account details (customer ID, store ID, store location, opened date, last transaction date (only reflects loyalty-linked purchases), and status) Campaign — campaign definitions (campaign ID, campaign start date, campaign end date, status) Campaign Member — member-campaign links (customer ID, campaign ID, opt_out_status, and response status)
Data Quality Issues Identified	**Issue 1 – No Cross-System Opt-out Communication** When a member opts out in one system, there is no mechanism to communicate that event to the other two systems. **Issue 2 – At-risk flag calculations** The at-risk member flag requires three components: the last purchase date from the Sales fact table; the last category purchased, derived from the Sales fact table joined to the Product dimension; and the last engagement date from the Contact History fact table. **Issue 3 – No unified Customer Identifier** Priya cannot determine if the same member unsubscribed across all channels or only one. No single source of truth links the Salesforce loyalty ID to the Shopify customer ID or the POS receipt number. ADR-002 required.
Volume and Velocity	**Initial load will require a full load:** 2.1 million Contact records average size ~3KB - 2 100 000 x 3KB = 6.3GB **Nightly batch:** CDC or delta extractions implemented. Estimated: 1-2% of members change daily. 2 100 000 x 2% = 41 000 records

	41 000 x 3KB = ~126MB per day for delta extractions I have gone for 2% as I like looking at the worst-case scenario to make sure we do not choke.
Integration Complexity	LOW-MEDIUM Reason 1: Extraction method known — ADF Salesforce connector v2 Bulk API 2.0. Well documented. Reason 2: No direct DB connection — Cloud API only. No production system impact. Reason 3: Identity resolution required — Salesforce loyalty ID must link to Shopify customer ID and POS receipt number. ADR-002 dependency adds Silver layer complexity.

8. Legacy SQL Server DW

System Overview	It is a Legacy SQL Server 2012 data warehouse. Business Owner: Johan van der Merwe Technical Owner: TBC - IT/DBA team. Currently serves the Finance domain, out of support since July 2022, and has a CDO decommission deadline of Q3 2026. There are two undocumented reports built in 2019. Nobody else knows they exist, but used monthly. Must be discovered and replicated before decommission.
Data Domains Served	Finance only.
Extraction Method	TBC — Pending discovery session with Johan van der Merwe. Approach: 1. Discovery — identify source systems and tables feeding the Legacy DW. 2. Report replication — rebuild Johan's two undocumented reports using the identified source data in the new platform. 3. Validation — two consecutive months parallel running. Both systems produce identical numbers. Johan signs off each month. 4. Decommission — Legacy DW switched off before Q3 2026 CDO deadline. Note: The Legacy DW is not a permanent source system for the new platform. It is being replaced. The actual upstream sources (likely SAP ECC) will feed the new platform directly.
Key Tables	Known: Two undocumented finance reports were built in 2019 that are used monthly by Johan's team. Unknown - TBC pending discovery: - What tables do the reports query? - What source systems feed those tables? - What business questions do the reports answer? - Are there other reports or objects Johan's team uses that have not been mentioned? Discovery action required: Sit with Johan and his team. Open both reports.

	Reverse engineer the SQL. Document every table, column, and calculation used. This must happen before any decommission work starts.
Data Quality Issues Identified	**Issue 1 — Out of Support** SQL Server 2012 is out of Microsoft enterprise support since July 2022. Running without the latest security patches creates vulnerability and legal compliance risk for a system holding financial data. **Issue 2 — Undocumented Reports** Two finance reports built in 2019 are used monthly by Johan's team. No documentation exists. Context and business questions answered by these reports are unknown. If the Legacy DW is decommissioned before these reports are discovered and replicated — the questions they answer cannot be reproduced. The full business impact is unknown until discovery is completed. **Issue 3 — Q3 2026 Decommission Deadline** If the Legacy DW is not decommissioned by Q3 2026 — the new platform and the Legacy DW will run simultaneously beyond the agreed deadline. This creates: - Two sources of truth for Finance — which number does Johan trust? - Ongoing maintenance cost for a system being replaced. - CDO programme governance breach — deadline formally agreed in BDR-001. - Johan's parallel running condition — two consecutive months of reconciliation must be completed BEFORE Q3 2026 deadline. The timeline is tight.
Volume and Velocity	TBC — Volume and velocity cannot be confirmed until the discovery session with Johan van der Merwe where we will identify the source systems and tables feeding the Legacy DW. Once discovered — volume will be assessed at the source level (likely SAP ECC) not at the Legacy DW level, as the new platform will extract from source systems directly.
Integration Complexity	MEDIUM Reason 1: Known technology — SQL Server 2012 is familiar. DBA expertise means we can query, reverse engineer, and replicate easily. Reason 2: Unknown content — Two mysterious undocumented reports. Context and business questions unknown until discovery with Johan. Reason 3: Tight decommission deadline — Q3 2026. Discovery, replication, two months parallel running, and sign-off must all complete before deadline. Time pressure adds complexity.

9. Excel Workbooks

System Overview	There are about 140 workbooks built over 11 years. Johan's Finance team uses them for the month-end close - 3 days' work. Business Owner: Johan van der Merwe Technical Owner: N/A - no technical owner, user-managed. Business rules trapped inside. Some rules are verbal agreements with the CFO - exists nowhere else.
Data Domains Served	Finance domain only.
Extraction Method	Manual documentation + dbt migration: Step 1: Sit with Johan's team. Open each workbook. Understand every formula. Document every business rule. Convert verbal CFO agreements to written rules. Step 2: Rebuild rules as dbt models. Visible. Version controlled. Auditable. No more Excel. Step 3: Validate the dbt output matches the Excel output for 2 consecutive months before Excel retires.
Key Tables	1. Month-end Close Pack 140 workbooks supporting the monthly close process. Currently takes Johan's team 3 days every month. 2. Creditors Aging Analysis Tracking outstanding invoices by supplier and age. 3. Three-way Invoice Matching Managing the 847-exception queue — invoices that failed PO + GRN + invoice match in SAP. Note: Exact workbook names and contents TBC — pending discovery session with Johan's team. All workbooks must be catalogued before any decommission work starts.
Data Quality Issues Identified	**Issue 1 — Business Rules Trapped in Excel** 140 Excel workbooks contain business logic that exists nowhere else. Some rules were verbal agreements with the CFO. These rules must be extracted, documented, and converted to dbt transformation models before Excel can be decommissioned. This is a process problem not a structural one — solvable with the right plan. **Issue 2 — Undocumented Knowledge** If the person who built a workbook leaves the company, they take their knowledge with them. The logic behind the formulas becomes impossible to maintain or trust without that person. **Issue 3 — No Version Control** Multiple people can update the same workbook simultaneously with no single source of truth. Johan cannot confirm which version contains the correct month-end figures, creating risk of reporting errors in the board presentation.
Volume and Velocity	Not applicable as a live data source. Excel workbooks are a business rule repository — not a transactional system. Volume and velocity apply to the dbt models that replace them, not to the workbooks themselves.
Integration Complexity	N/A - not a transactional system but rather business rules being migrated to dbt.

10. Salesforce Marketing Cloud

System Overview	It is an email and campaign execution platform. Business Owner: Rethabile Sithole. Technical Owner: TBC - IT team. Rethabile's team uses it to send campaigns to loyalty members. Links to Salesforce CRM for member targeting. It serves the Marketing domain.
Data Domains Served	Marketing
Extraction Method	Method: Marketing Cloud automated SFTP export. Tool: Azure Data Factory picks up files from SFTP. Format: CSV files Schedule: Nightly batch — after campaign activity settles.
Key Tables and Entities	Campaign Sends: Campaign_id, name, date_posted Email Opens: customer_email, customer_firstname, customer_lastname, campaign_id, campaign_name Link Clicks: clicked_id, campaign_id, click_date, click_count Unsubscribes: customer_id, customer_firstname, customer_lastname, unsubscribe_reason, unsubscribe_date Bounce Rates: campaign_id, customer_email, bounce_count
Data Quality Issues Identified	**Issue 1 – Opt-out problem** all three marketing systems, Marketing Cloud, Meta Business Manager, Google Analytics 4, and Salesforce CRM, must be updated within 4 hours after a user has unsubscribed. **Issue 2 – Meta Data Delay Impact** Meta confirms ad delivery 24-48 hours after the ad is served. When Marketing Cloud checks the Contact History fact table before sending an email — Meta contact events from the previous day are missing. The contact frequency count is understated. Marketing Cloud may send an email believing a member has received 2 contacts this month when they have actually received 3 via Meta. This risks breaching the 4-contact monthly cap and triggering a POPIA compliance breach. Solution: Optimistic approach agreed with Priya — use confirmed counts and alert same day when Meta data reveals a breach. FR-CUS-001.
Volume and Velocity	TBC — Volume and velocity to be confirmed with Rethabile Sithole during Step 5 technical investigation. Information required:

	- Number of campaigns per week - Average members targeted per campaign - Estimated event volume per day (sends, opens, clicks, unsubscribes) Until confirmed — Marketing Cloud SFTP exports are expected to be significantly smaller than POS or SAP volumes. Campaign event data is lightweight per record. Estimated ~10-50MB per day pending Rethabile confirmation.
Integration Complexity	MEDIUM Reason 1: Extraction method known — SFTP export picked up by ADF. Reduces uncertainty. Reason 2: Unknown data structure — Exact tables, columns, and export format TBC pending Marketing Cloud access. Adds some uncertainty. Reason 3: No direct DB connection — SFTP file pickup only. No production system impact.

11. Meta Business Manager

System Overview	It is a paid advertising platform for Facebook and Instagram. Business Owner: Rethabile Sithole Technical Owner: TBC — IT/Marketing team It serves the Marketing team and is used for paid social campaigns targeting loyalty members and new customers. The current challenge - Ad delivery data arrives 24-48 hours late. Creates the contact frequency counting problem we documented with Priya.
Data Domains Served	Marketing
Extraction Method	Method: Meta Marketing API via ADF HTTP connector. Key consideration: Meta data arrives 24-48 hours late. Pipeline must account for this delay. Do not expect real-time data. Design for daily batch with known latency window.
Key Tables and Entities	Campaign Performance: campaign_id, campaign_name, date_posted, status Ad Spend: campaign_id, ad_id, spend_amount, currency, date Impressions: campaign_id, ad_id, impressions_count, reach, date Link Clicks: clicked_id, campaign_id, click_date, click_count

	Note: All data arrives 24-48 hours late. Date fields reflect when the ad was served — not when Meta confirms the data.
Data Quality Issues Identified	**Issue 1 — Late Arriving Data** Meta confirms ad delivery 24-48 hours after the ad is served. The Contact History fact table will undercount member contacts until Meta data arrives. This risks breaching the 4-contact monthly cap — a POPIA compliance risk. Solution: Optimistic approach agreed with Priya — use confirmed counts and send a breach alert the same day Meta data confirms an overage. FR-CUS-001 and FR-CUS-002. **Issue 2 — No Marketing Attribution** Source of Truth Meta uses its own attribution model which conflicts with Salesforce Marketing Cloud and GA4 attribution. Double and triple counting of conversions is likely across the three systems. ADR-006 must be formally agreed before any Gold layer marketing attribution build. BPA-FINDING-002.
Volume and Velocity	TBC — Volume and velocity to be confirmed with Rethabile Sithole during Step 5 technical investigation. Information required: - Number of campaigns per week - Average members targeted per campaign - Estimated daily event volume (impressions, clicks, ad spend) Until confirmed — Meta API data is expected to be smaller than POS or SAP volumes. Ad performance data is lightweight per record. Estimated ~10-50MB per day pending Rethabile confirmation. Note: All data arrives 24-48 hours late. Volume estimates reflect confirmed data not real-time.
Integration Complexity	MEDIUM Reason 1: Extraction method known — Meta Marketing API via ADF HTTP connector. Reduces technical uncertainty. Reason 2: No direct DB connection — API only. No Meta production system performance impact. Reason 3: Data structure TBC — Exact fields and objects to be confirmed during technical investigation with Rethabile Sithole. Adds some uncertainty.

12. Google Analytics 4

System Overview	GA4 is a web analytics platform that tracks customer behaviour on RetailSphere's Shopify online store. Business Owner: Rethabile Sithole Technical Owner: TBC — IT/Marketing team It captures website visits, page views, product views, add to cart events, purchase conversions, traffic sources, and it serves the Marketing domain. It matters because Rethabile and team needs it for attribution, understanding which marketing channels drive online conversions. ADR-006 dependency.
Data Domains Served	Marketing

Extraction Method	Method: GA4 Data API via ADF HTTP connector. Schedule: Nightly batch. GA4 data typically available next day — not real-time. Key consideration: GA4 data has a processing delay of 24-48 hours — similar to Meta. Design for daily batch with known latency window. Authentication: Google OAuth 2.0 required. Service account credentials needed for ADF connection. IT team to provision access.
Key Tables	Product Views product_id, product_name, number_of_views, number_of_purchase Product Add to cart events product_id, customer_id, cart_event, event_date Traffic Sources campaign_id, campaign_name, date_posted, status Purchase Conversion customer_id, customer_name, email, campaign_id, campaign_name, purchase date, amount, quantity.
Data Quality Issues Identified	**Issue 1 - Late-arriving data** GA4 data has a 24-48 hour processing delay — similar to Meta. Design for a daily batch with a known latency window. The Contact History fact table will undercount member contacts until GA4 data arrives. This has the same risk as Meta with the risk of breaching the 4-contact monthly cap - a POPIA compliance risk. **Issue 2 — No Common Campaign Identifier Salesforce uses loyalty ID.** Meta uses pixel-based browser tracking. GA4 uses client ID. Three different identifiers for the same person. Attribution across systems requires identity resolution first
Volume and Velocity	TBC — Volume and velocity to be confirmed with Rethabile Sithole during Step 5 technical investigation. Information required: -Number of Purchase Conversions -Campaign performance — which campaigns are driving traffic and conversion Until confirmed, GA4 data is expected to be smaller than POS or SAP volume. Note: All data arrives 24-48 hours late. Volume estimates reflect confirmed data not real-time.
Integration Complexity	MEDIUM Reason 1: Extraction method known — GA4 Data API via ADF HTTP connector. Reduces technical uncertainty. Reason 2: No direct DB connection — API only. No GA4 production system performance impact. Reason 3: Data structure TBC — Exact fields and objects to be confirmed during technical investigation with Rethabile Sithole. Adds some uncertainty.

13. Supplier Contract PDFs

System Overview	There are 340 unstructured PDF documents. Stored on a shared drive. Contain: contracted rates, lead times, payment terms. Business Owner: Sipho Dlamini Technical Owner: N/A — unstructured documents on shared drive. Serves the Procurement domain as these contracts are handled by Sipho. No database, no API. Sipho needs this for FR-PRO-003 price variance calculation. ADR-003 must be resolved first.
Data Domains Served	Procurement
Extraction Method	Method: LLM-assisted hybrid approach. Stage 1 — Text Extraction: pypdf extracts raw text from each PDF document. Handles standard PDF text layers. Stage 2 — LLM Structuring: LLM (Claude or similar) reads the extracted text and identifies key fields: - Contracted rate per product - Lead time in days - Payment terms (30/60/90 days) - Supplier name and code - Contract start and end date Outputs structured JSON or CSV ready for Bronze layer loading. Stage 3 — Human Validation Queue: Low confidence extractions flagged for Sipho's team to review and confirm before loading to Bronze. Prevents incorrect contracted rates entering the price variance calculation. Dependency: ADR-003 must be formally agreed and signed before this extraction approach is built.
Key Tables and Entities	- Contracted rate per product - Lead time in days - Payment terms (30/60/90 days) - Supplier name and code - Contract start and end date
Data Quality Issues Identified	**Issue 1** - No text layer extracted, due to different PDF formats, scanned documents or images. **Solution**: OCR (Optical Character Recognition) as a fallback for scanned PDFs. Tools: Azure Form Recogniser or Tesseract OCR can handle scanned documents that pypdf cannot read. **Issue 2** - When the LLM extracts a contracted rate incorrectly could lead to wrong numbers in the report that would cause Sipho to make a wrong decision regarding supplier rates, as he won't be able to compare competitive versus what the market offers. **Solution:** Human validation queue. Low-confidence LLM extractions flagged for Sipho's team review before loading to Bronze. Stage 3 of the hybrid approach.
Volume and Velocity	Not Applicable, as it will be a one-time load. Minimal ongoing impact in terms of data growth.
Integration Complexity	MEDIUM-HIGH Reason 1: Extraction method known — LLM-assisted hybrid approach. pypdf + LLM + human validation.

	Reason 2: Not yet tested or approved — ADR-003 must be formally agreed before build begins. Reason 3: Human validation process not yet established — who approves, what threshold, what workflow. Reason 4: Ongoing maintenance manual — new or updated contracts must be manually processed through the same three stages with no exceptions. No automated detection of new contracts on the shared drive.

14. Integration Complexity Summary

Source System	Domain	Complexity	Primary Challenge	Priority
SAP ECC 6.0	Procurement, Inventory, Finance	MEDIUM	Single point of failure across three domains.	HIGH — extract first. Three domains depend on it.
Oracle MICROS POS	Sales, Inventory	HIGH	87 stores, 87 different schemas. CDC feasibility TBC.	CRITICAL — most complex. Start investigation immediately.
Shopify Plus	Sales	LOW-MEDIUM	ADR-001 and ADR-002 identity resolution required.	MEDIUM — well documented API.
Salesforce CRM	Customer Loyalty, Sales	LOW-MEDIUM	ADR-002 identity resolution. Master customer record.	HIGH — ADR-002 depends on it.
Legacy SQL Server DW	Finance	MEDIUM	Two undocumented reports. Q3 2026 decommission deadline.	HIGH — discovery urgent. Deadline is tight.
Excel Workbooks	Finance	N/A	Business rules trapped. Manual migration to dbt.	HIGH — must complete before Excel decommission.
Salesforce Marketing Cloud	Marketing	MEDIUM	Data structure TBC. SFTP export format to confirm.	MEDIUM — ADR-006 must be agreed first.
Meta Business Manager	Marketing	MEDIUM	24–48-hour data delay. Attribution complexity.	MEDIUM — ADR-006 must be agreed first.
Google Analytics 4	Marketing	MEDIUM	24–48-hour data delay. Identity resolution for conversion attribution.	MEDIUM — ADR-006 must be agreed first.
Supplier Contract PDFs	Procurement	MEDIUM-HIGH	Unstructured documents. LLM extraction. Manual validation process TBC.	HIGH — ADR-003 must be agreed first.

15. Next Steps

- CDO review and sign-off on SSI-001 v1.0.
- IT team engagement — confirm technical ownership and access for all source systems marked TBC.
- Oracle MICROS POS investigation — confirm DBMS, network connectivity, and CDC feasibility across all 87 stores. Highest priority.
- Legacy SQL Server DW discovery — sit with Johan van der Merwe to identify and document the two undocumented reports before Q3 2026 decommission deadline.
- ADR-003 — Supplier Contract PDF Extraction strategy formally agreed and signed before build.
- ADR-006 — Marketing Attribution Logic formally agreed and signed before any Marketing mart build.
- Rethabile Sithole to confirm Marketing Cloud, Meta, and GA4 data volumes and export formats.
- Proceed to Step 6 — Risk Analysis and ADR resolution.
- Upload SSI-001 to GitHub: docs/05-source-system-investigation/ SSI-001_v1.0.docx

16. Document Change Log

Version	Date	Author	Description
1.0	28 May 2026	T. Mbalo	Initial release. 10 source systems investigated across 6 business domains. Integration complexity ranked and prioritised. Pending CDO sign-off.

— End of Document —